LEVEL 2 PROGRAMS

PRACTICE PROGRAMS -JAVA PROGRAMMING:

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Matrix multiplication

Your Code

```java
public class Main {
    public static void main(String[] args) {
        int[][] a = {
            {1, 2},
            {3, 4}
        };
        int[][] b = {
            {5, 6},
            {7, 8}
        };
        int r1 = a.length;
        int c1 = a[0].length;
        int r2 = b.length;
        int c2 = b[0].length;
        if (c1 != r2) {
            System.out.println("Matrix multiplication not possible.");
            return;
        }

        int[][] result = new int[r1][c2];
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                result[i][j] = 0;
                for (int k = 0; k < c1; k++) {
                    result[i][j] += a[i][k] * b[k][j];
                }
            }
```

Input

stdin input

Output

```
19 22
43 50
```

```java
            System.out.println("Matrix multiplication not possible.");
            return;
        }

        int[][] result = new int[r1][c2];
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                result[i][j] = 0;
                for (int k = 0; k < c1; k++) {
                    result[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                System.out.print(result[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Q2 . Largest number

10 marks

. Largest number

Your Code

```java
public class main{
    public static void main(String[] args){
        int[] arr={12,45,67,23,89,34};
        if(arr.length==0){
            System.out.println("array is empty");
            return ;
        }
        int max = arr[0];
        for(int i =1;i<arr.length;i++){
            if(arr[i]>max){
                max=arr[i];
            }
        }
        System.out.println("largest element :"+max);
    }
}
```

Run Save ✓ Draft saved

Input

stdin input

Output

```
largest element :89
```

Q3 Smallest number

10 marks

Smallest number

Your Code

```java
public class Main {
    public static void main(String[] args) {
        int[] numbers = {15, 42, 7, 89, 23, 4};
        if (numbers.length == 0) {
            System.out.println("Array is empty");
            return;
        }
        int small = numbers[0];
        for (int i = 1; i < numbers.length; i++) {
            if (numbers[i] < small) {
                small = numbers[i];
            }
        }
        System.out.println("Smallest element: " + small);
    }
}
```

Run Save

Input

stdin input

Output

```
Smallest element: 4
```

Q4 . Nth largest number

10 marks

. Nth largest number

Your Code

```java
public class NthLargest {
    public static void main(String[] args) {
        int[] arr = {14, 67, 23, 89, 45, 90, 12};
        int n = 2;
        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] < arr[j]) {
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("The " + n + "th largest number is: " + arr[
    }
}
```

Input

```
stdin input
```

Output

```
The 2th largest number is: 89
```

Q5 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```java
public class PalindromeCheck {
    public static void main(String[] args) {
        String str = "malayalam";
        String rev = "";
        for (int i = str.length() - 1; i >= 0; i--) {
            rev = rev + str.charAt(i);
        }
        if (str.equalsIgnoreCase(rev)) {
            System.out.println(str + " is a Palindrome");
        } else {
            System.out.println(str + " is Not a Palindrome");
        }
    }
}
```

Input

```
stdin input
```

Output

```
malayalam is a Palindrome
```

Run Save

Q6 **Removing duplicates in array** 10 marks

Removing duplicates in array

Your Code

```java
public class RemoveDulplicates{
    public static void main(String[] args){
        int[] arr={1,2,2,3,4,4,5};
        int length=arr.length;
        System.out.print("original array: ");
        for(int num : arr){
            System.out.print(num + " ");
        }
        System.out.print("\nArray without duplicates :");
        for(int i=0;i<length;i++){
            boolean isDuplicate = false;
            for(int j =0;j<i;j++){
                if(arr[i] == arr[j]){
                    isDuplicate = true;
                    break;
                }
            }
            if(!isDuplicate){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

Input

stdin input

Output

```
original array: 1 2 2 3 4 4 5
Array without duplicates :1 2 3 4 5
```